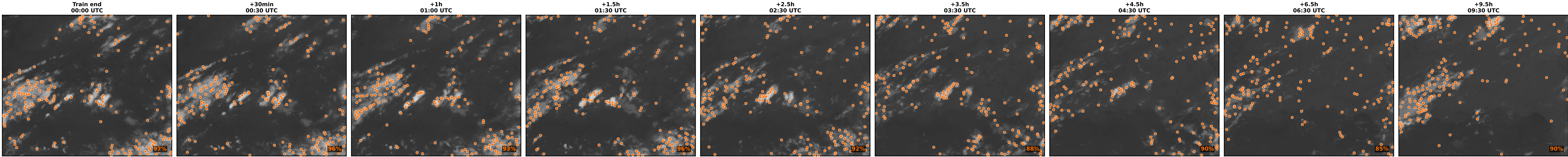


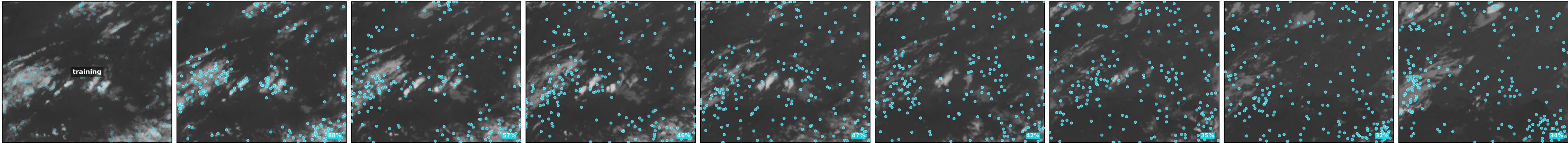
GOES-16 Cloud Forecasting: 30-min Cadence, Raw Coordinates
N=R=200 | Per-frame gaussian_iso | 3-cycle training | On-cloud % per frame

$\rho=0.9, \alpha=0.9, \lambda=0.01$ | Velocity wins +60.6% L² over Positions

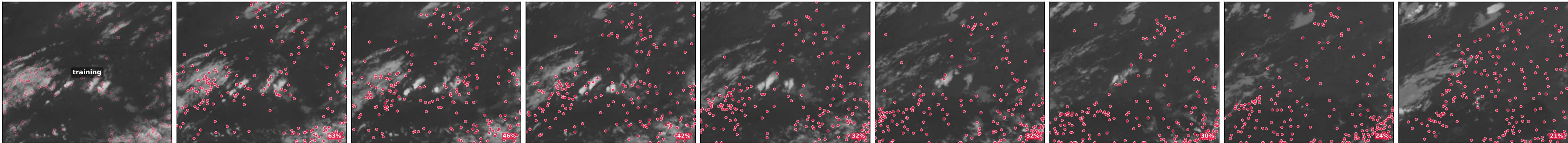
Ground
Truth



Velocity
Forecast



Positions
Forecast



True cloud particles (N=200) Velocity forecast (R=200) Positions forecast (R=200)